

ABHISHEK AKARAM PATIL

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PERSONAL STATEMENT

Geospatial consultant with 7+ years of experience, advising government, water, infrastructure, education, and telecom organisations on spatial strategy, workflow transformation, and geospatial product development — streamlining their desktop & field operations to make work faster, more accurate, & just a bit more exciting than endless spreadsheets & manual mapping. Because who doesn't love saving time & looking good on a dashboard? Deep technical expertise in remote sensing, spatial analytics, and automation with the ability to engage senior stakeholders and advise on geospatial roadmaps, policy, and program delivery.

KEY ACHIEVEMENTS

- Chartered Geographer (GIS), Royal Geographical Society (RGS), UK
- Presented at Geo-Business, London 2026. Finalist: Construction Computing Awards 2026 (GIS Project of the year)
- Recognised as a 'Top Geographical Information Systems (GIS) Voice' by LinkedIn
- Appointed on a panel of experts for Geography subject at Amity University, Noida, India. (2023)
- Recipient of [NESA UGRAD](#), a fully funded scholarship by the U.S. govt. (One of the 5 students from India)

PROFESSIONAL EXPERIENCE

GIS Analyst @One Alliance

Skanska, UK | Anglian Water, UK

Oct 2025 – Present

- Led the implementation of scalable geospatial data engineering capabilities using **GeoParquet, Apache Sedona, and Databricks, addressing high-volume spatial analytics, and distributed geoprocessing workloads** that were not feasible within existing Esri-centric architectures.
- Provided geospatial advisory to Planning, Design, and Engineering stakeholders to **embed 3D visualisation, VR** and spatial analysis into infrastructure planning, scenario evaluation, and strategic decision-making workflows.
- Designed and developed an **Alliance Search Engine** to improve the visibility and accessibility of data across multiple platforms, reducing duplication of effort and enhancing operational efficiency
- As a technical lead, actively **promoted adoption** of GIS, Remote Sensing, and Spatial Intelligence across the One Alliance partnership — advising senior stakeholders at **Anglian Water, Skanska, MWH, Binnies, Mott MacDonald, Sweco, and SPA** on geospatial strategy, workflow transformation, and spatial innovation opportunities.
- Designed and implemented custom **Python and Arcade-based automation tools** to streamline repetitive geospatial workflows, data sharing, significantly reducing processing time and enhancing data quality and operational efficiency.
- Led strategic development and cross-partner advisory for an **Early-Stage Operational Risk Intelligence Platform** — consulting with alliance partners to define requirements, validate methodology, and deliver from **MVP** to enterprise rollout, supporting mitigation of approximately £6M in operational losses
- As an **Analytics & technical lead**, supported a **national urgent review of buildings at risk from RAAC**. Led the development of methodologies to identify flat and sloped roofs across the AW region using **LiDAR** point clouds and **Sentinel-1** imagery to map black-roof surfaces.
- Consulted with Solutions teams and field operations partners to design and deploy enterprise geospatial applications using **Survey123, Experience Builder, Field Maps, and 360° imagery** — enabling regional utility infrastructure assessments across the alliance (e.g. **Asset: HealthScape**).
- Consulted with **Planning and Environmental stakeholders** across alliance partners to design and deliver the BERA platform — translating complex multi-stakeholder requirements into a spatial solution analysing 500+ sites against 100+ constraints, cutting assessment timelines from month to minute and saving ~£70k per cycle.
- Advised on an **automated enterprise Environmental Risk Assessment workflow** — consolidating 100+ geospatial datasets from multiple partner organisations into rapid spatial screening and risk / environmental constraint-classification tools with instant report-ready outputs.
- Driving **CAD/BIM (GeoBIM)** integration advisory across design, engineering, and operational teams within the alliance.

GIS Specialist (Smart Rivers, Environmental Solutions)

Wessex Water, YTL, UK

Dec 2024 – Oct 2025

- Led initiatives to embed GIS-based database management into core workflows across the Environmental Solutions team (100+ staff), promoting **GIS awareness and adoption** across disciplines. Acted as a key liaison between technical and non-technical teams to translate complex geospatial data into actionable decisions with **ESRI tools**.
- Tailored GIS solutions for projects in **Land Entry, Ecology, Planning, & Environmental Investigations**.
- Developed **spatial digital workflows** with Python, ESRI tools, FME & SQL to support the deployment & management

of approximately 2,000 sensors across 1,200 sites as part of the Section 82 CWQM program.

- Developed **GIS dashboards** using **Experience Builder, Image Server, Utility Network, Mobile Apps (Survey123, Field Maps)** --enabling spatial data visualisation, editing, analysis, and database management.
- Planned & processed **USV & UAV surveys (photogrammetry)** (Processing time reduced by 40%)
- Developed a workflow for designing and building a **360-degree viewer** for monitoring sites & assets. Cost reduced by GBP 20,000/Year
- Collaborated with cross-functional teams to integrate datasets, streamline reporting processes, **develop a project tracker**, & enable real-time water quality **data visualisation for public-facing dashboards**.
- Upskilling multiple teams with GIS knowledge via hands-on training and workshops.

Geospatial Technician – Civil Servant, Land Mapping Team

Director-General Net Zero, Scottish Government

Sept 2024 – Dec 2024

- Developed expertise in land mapping & case management using **bespoke GIS software LIPIS & aerial photography**.
- Handled complex cases and **customer service enquiries**, ensuring efficient delivery of rural support services.

GIS & Remote Sensing Expert - Network Planner:

Environmental Management & Planning Dept., AMN Ltd., Milton Keynes, England, UK

Sept 2023 – Sept 2024

UK based AMN - builds, owns, operates and maintains mobile network infrastructure in Africa and Latin America.

- Led the design and deployment of **GIS-enabled planning systems** for mobile network expansion across Africa and Latin America—serving both internal stakeholders and underserved communities. Transformed company's manual planning into a comprehensive GIS-based planning system using, increasing efficiency by 25%.
- Optimised **spatial data management & analysis** processes by introducing efficient **ETL workflows** that reduced processing time by 75%. Delivered actionable insights through geospatial analysis.
- Collaborated with country managers, enabling them to troubleshoot, adopt, and scale automated workflows.

Assistant Professor of Geography:

PES's Modern College of Arts, Science, and Commerce College, Pune, India

Jul 2019 - May 2022

- **Delivered 21 hours of weekly lectures** on Geography, Cartography, Environment, GIS and Remote Sensing.
- Led **student-driven geospatial initiatives** in collaboration with NGOs and civic groups, using open-source tools.

RESEARCH PUBLICATION

- **Assessing peat surface motion using Interferometric Synthetic Aperture Radar (InSAR) in the Great Fen area of Cambridgeshire, UK** (in collaboration with Cranfield University & University of Nottingham) 2026

EDUCATION

Self-MBA Program by MBATHinking	Ongoing
NASA Applied Remote Sensing (ARSET) training: Monitoring Groundwater Changes	2026
Data-driven approaches to hydrological science, UKCEH Summer School	2025
MSc Advanced GIS & Remote Sensing, Cranfield University, U.K.	2022-23
MSc Geography, Lovely Professional University, India (with University Academic Honor)	2017-19
Bachelor in Geography, Mumbai University, India	2012-16
Humboldt State University, California, USA	2014-15

CERTIFICATION

- BIM-GIS Integration for Architecture, Engineering & Construction by ESRI Sept 2026
- Scrum Master Certification by Knowledge Academy Jan 2026
- Generative AI Mastermind by Outskill Aug 2025
- FME (Feature Manipulation Engine) Essentials by Udemy July 2025
- AutoCAD Map 3D Essential Training Jun 2025

Also Exploring mountains, forests, —and how AI and ML might someday navigate them better than I do